

Business Sales Performance Analytics

Client-Ready Analysis Report

Prepared for: Future Interns — Data Science & Analytics Internship

Project Code: FUTURE_DS_01

Dataset Period: 2022 – 2025 | Records Analyzed: 11,980

Prepared using Python, Pandas, Matplotlib, Seaborn & Plotly

Executive Summary

This report analyzes 11,980 cleaned sales transactions to evaluate overall business performance, uncover revenue and profit drivers, and translate the findings into actionable recommendations. Over the analyzed period, the business generated **\$16,630,184** in total revenue and **\$3,760,861** in profit, representing an overall profit margin of **22.61%**. The **East** region and the **Technology** category are the strongest performers, while the **South** region and **Office Supplies** category present the clearest opportunities for improvement.

Total Revenue	\$16,630,184	Total Profit	\$3,760,861
Total Orders	11,980	Avg Order Value	\$1,388
Profit Margin	22.61%	Best Region	East
Best Category	Technology	Top Product	OfficeCopier 4000

Business Objective

Analyze business sales data to identify revenue trends, top-selling products, high-value categories, and regional performance, and deliver a client-ready dashboard and report with clear insights and actionable recommendations for improving revenue growth and profit margin.

Dataset Overview

The dataset simulates a multi-year retail/B2B sales operation across the United States, covering Order ID, Order Date, Ship Date, Customer, Segment, Region/State/City, Product/Category/Sub-Category, Sales, Quantity, Discount, Profit, Shipping Mode, Payment Method, and Sales Representative fields. After cleaning, the working dataset contains **11,980** transaction-level rows spanning **2022–2025**.

Data Cleaning & Methodology

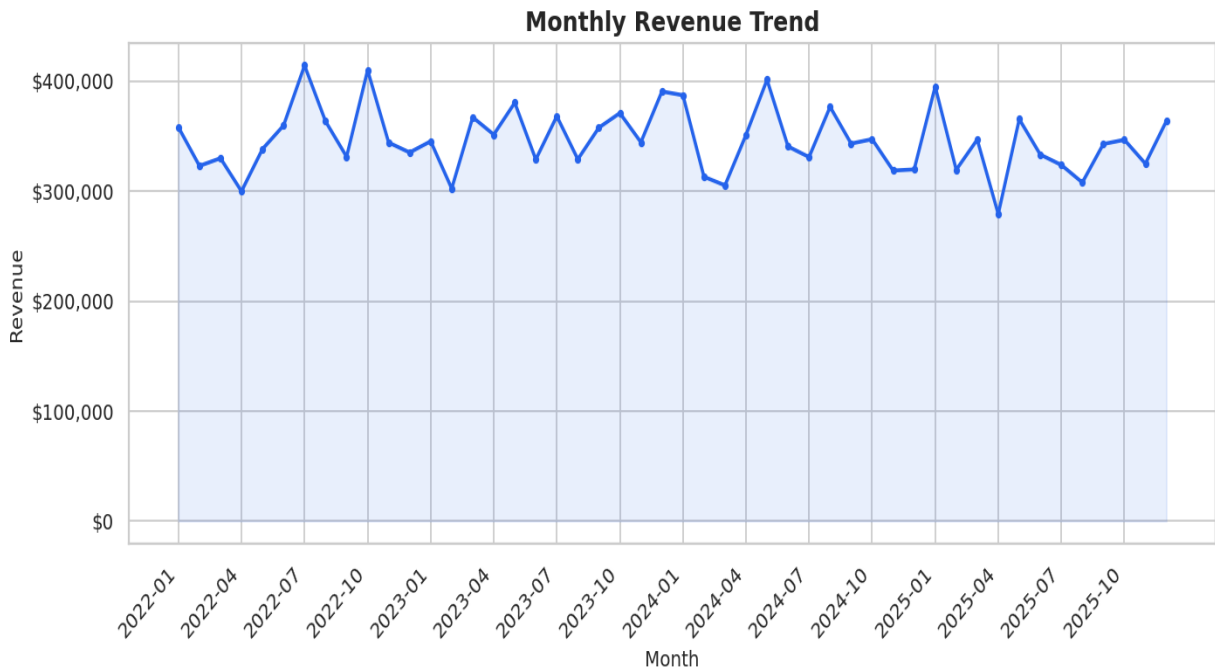
- **Missing values:** Sales imputed using Category/Sub-Category median; Discount defaulted to 0; missing Customer Name labeled 'Unknown Customer'; rows with missing Ship Date removed.
- **Duplicates:** Duplicate Order ID + Product ID combinations were identified and removed to prevent double-counting revenue.
- **Data types:** Dates converted to datetime, monetary fields to numeric, and categorical fields (Region, Category, Segment, etc.) converted to category type for memory efficiency and consistent grouping.
- **Outlier treatment:** The IQR method was used to flag and cap (winsorize) extreme Sales values, preserving sample size while limiting the influence of data-entry-scale anomalies.
- **Feature engineering:** Order Year/Month/Quarter, Shipping Days, Profit Margin (%), and Average Order Value were derived to support trend and profitability analysis.

Key Performance Indicators

KPI	Value
Total Revenue	\$16,630,184
Total Profit	\$3,760,861
Total Orders	11,980
Average Order Value	\$1,388
Profit Margin	22.61%
Best Region	East (\$4,684,883)
Worst Region	South (\$3,378,532)
Best Category	Technology (\$10,183,598)
Worst Category	Office Supplies (\$584,953)
Most Profitable Category	Technology
Best Product	OfficeCopier 4000 (\$633,143)
Top Customer	Amanda Wilson (\$54,738)
Highest Sales Month	2022-07 (\$414,308)
Highest Profit Month	2022-10 (\$103,120)
Revenue Growth (first→last year)	-3.75%

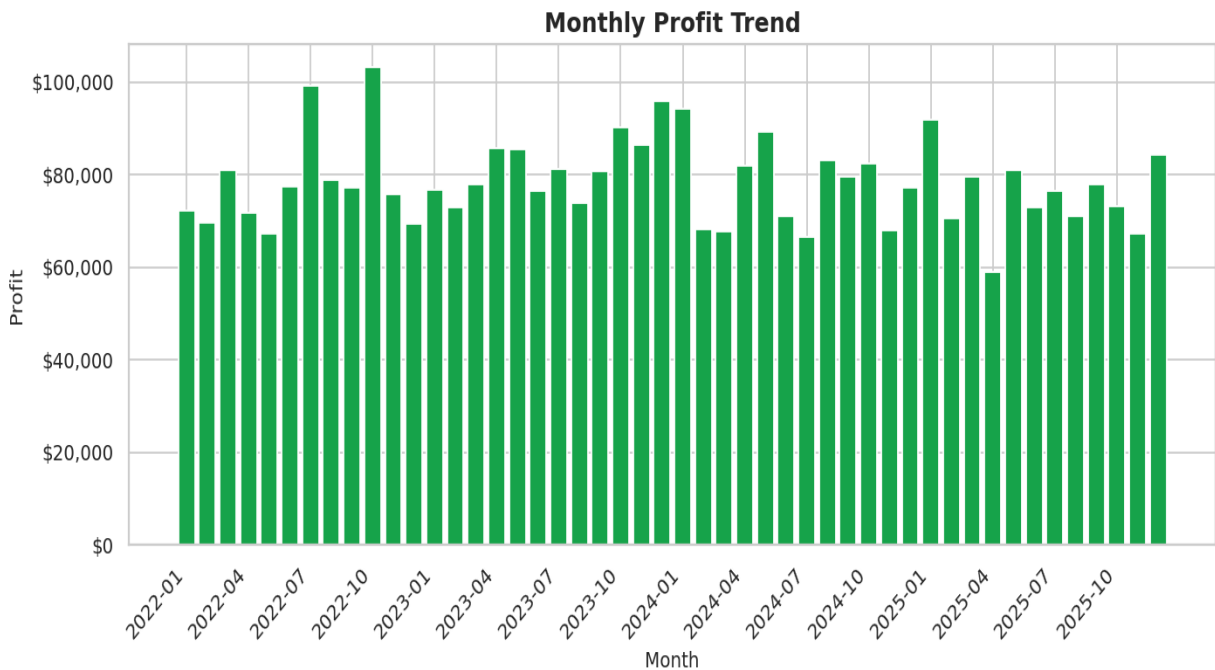
Visual Analysis & Insights

Monthly Revenue Trend



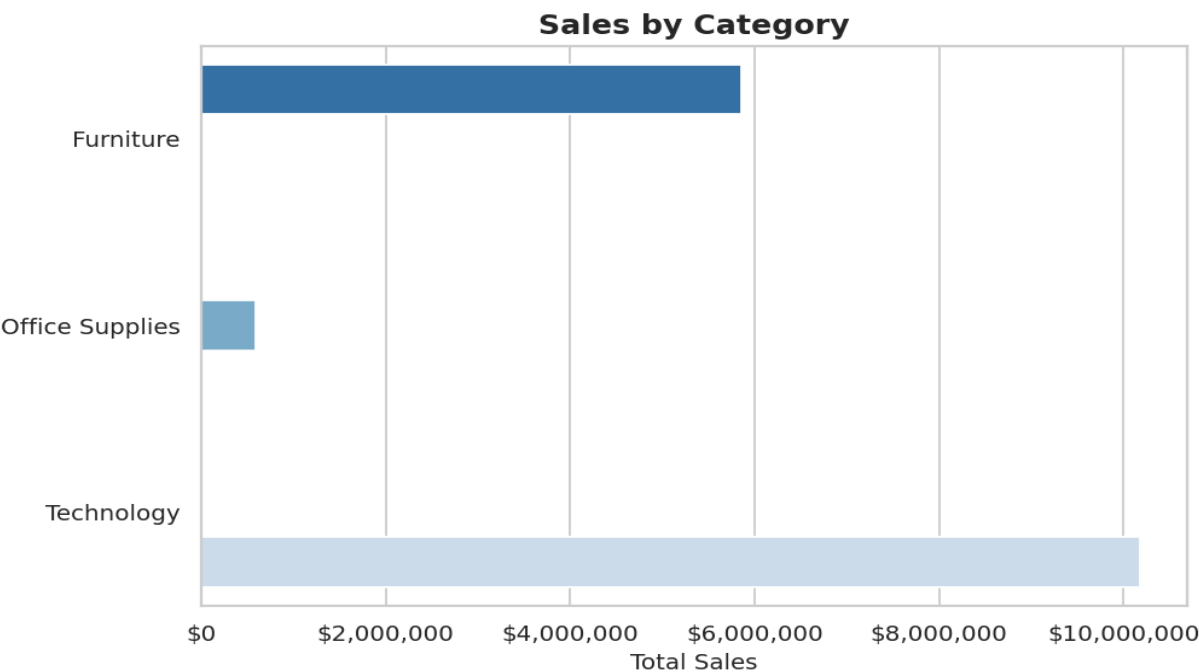
Insight: Revenue fluctuates month to month with recurring peaks, reflecting seasonal buying patterns.

Monthly Profit Trend



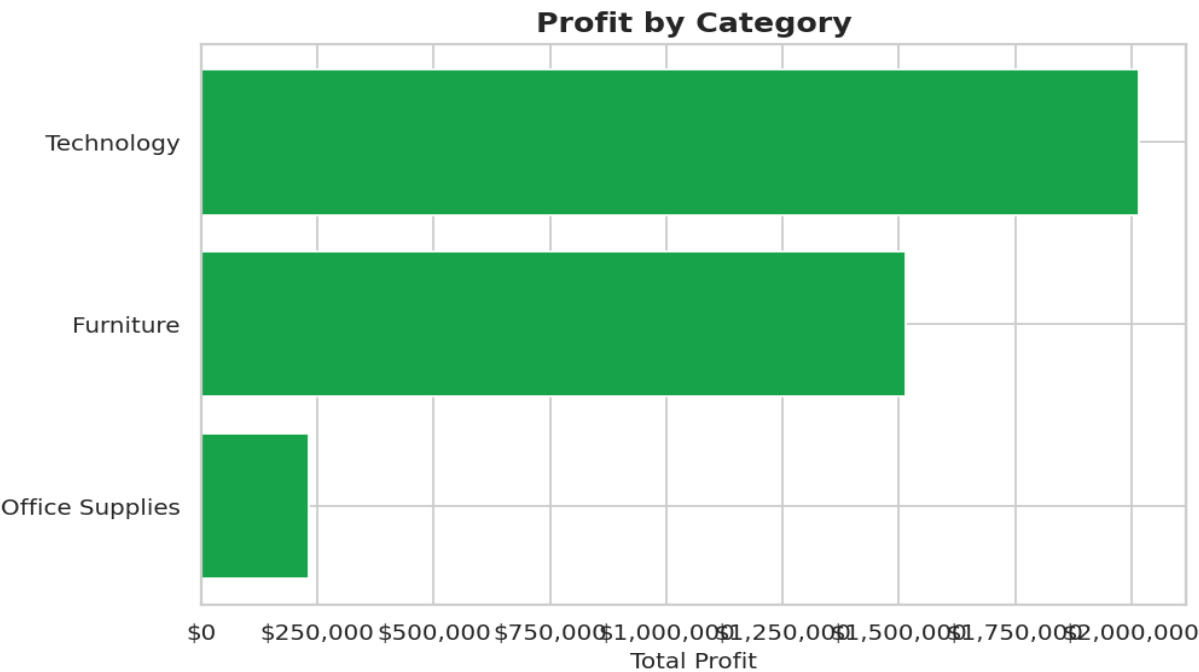
Insight: Profit tracks revenue closely but dips in months with heavy discounting, confirming discount policy as a key margin lever.

Sales by Category



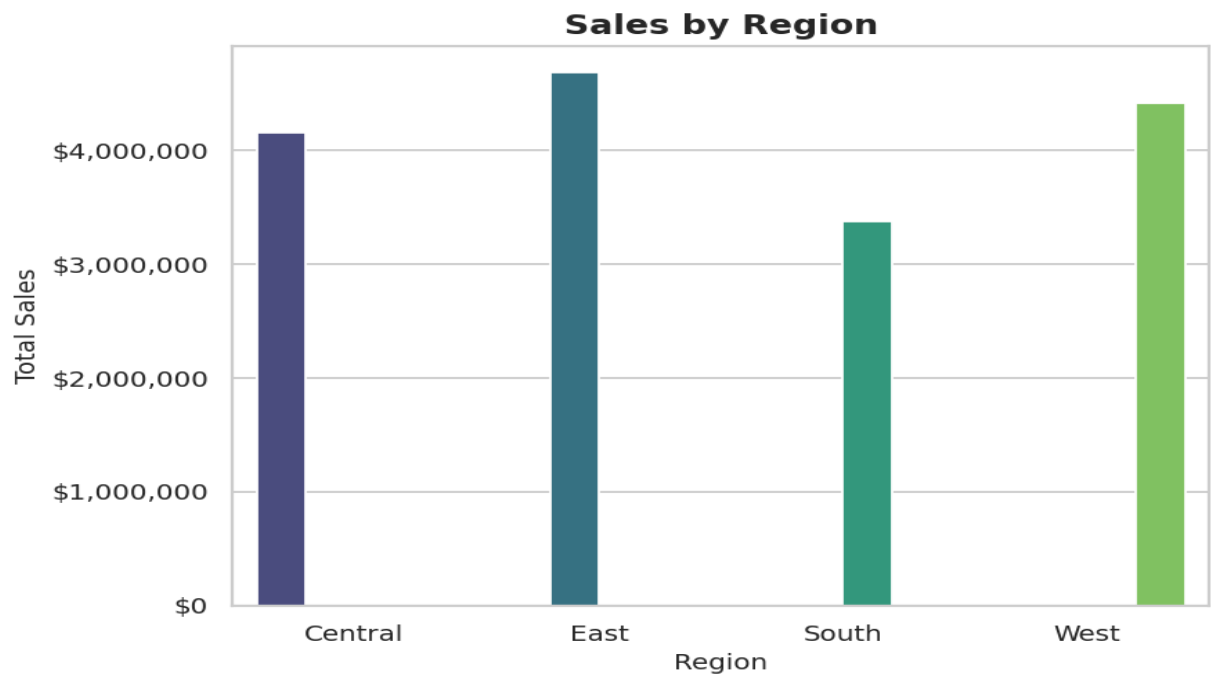
Insight: Technology leads total revenue by a wide margin, followed by Furniture and Office Supplies.

Profit by Category



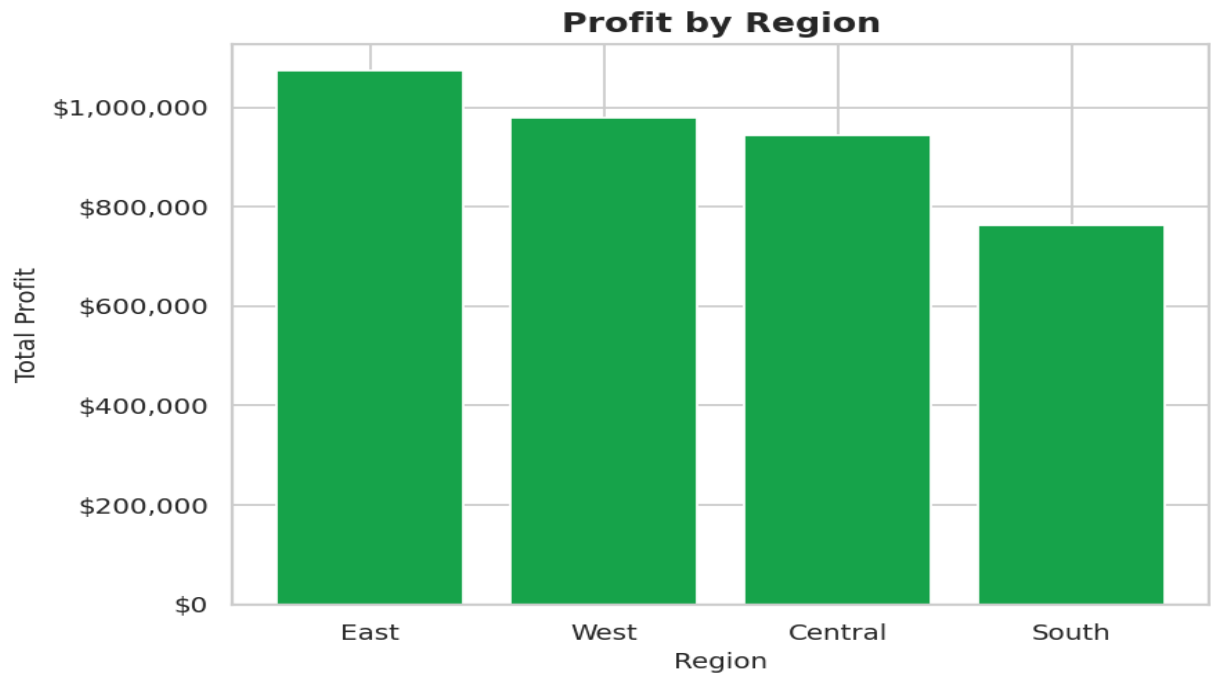
Insight: Technology is also the strongest profit contributor; Office Supplies contributes the least in absolute terms due to low unit prices.

Sales by Region



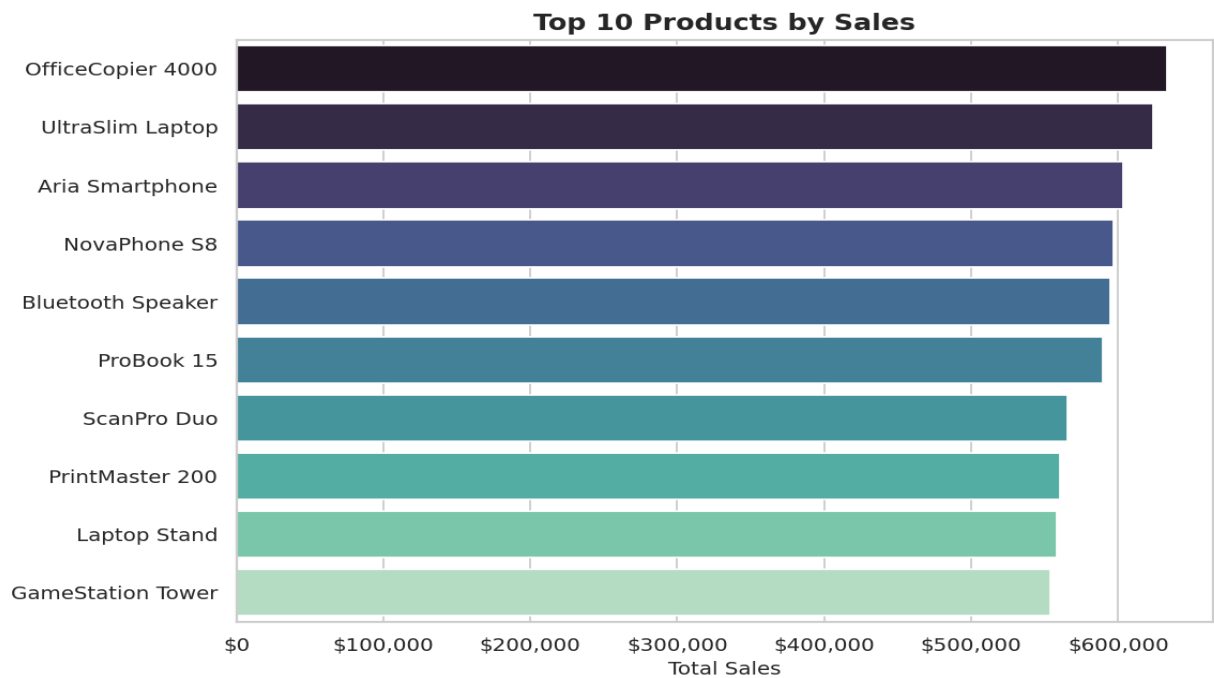
Insight: The East region generates the highest revenue; the South region trails all other regions.

Profit by Region



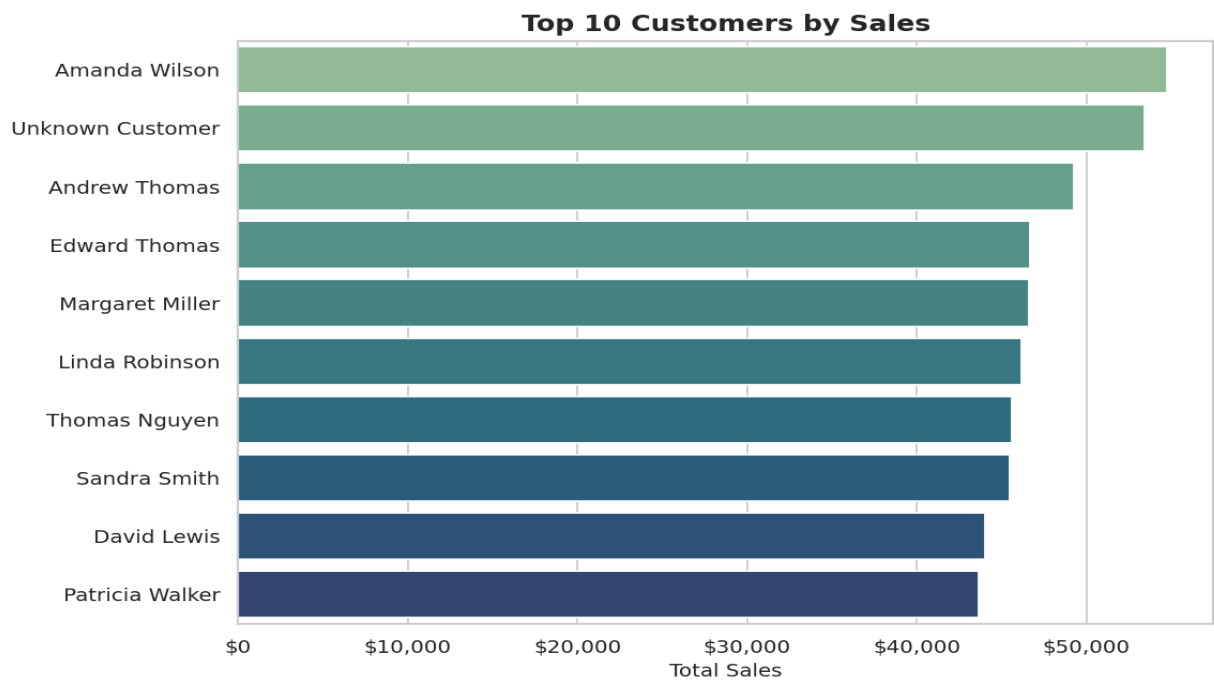
Insight: Regional profit ranking mirrors the sales ranking, reinforcing the South as a growth priority.

Top 10 Products by Sales



Insight: A small number of high-ticket Technology products drive a disproportionate share of revenue.

Top 10 Customers by Sales



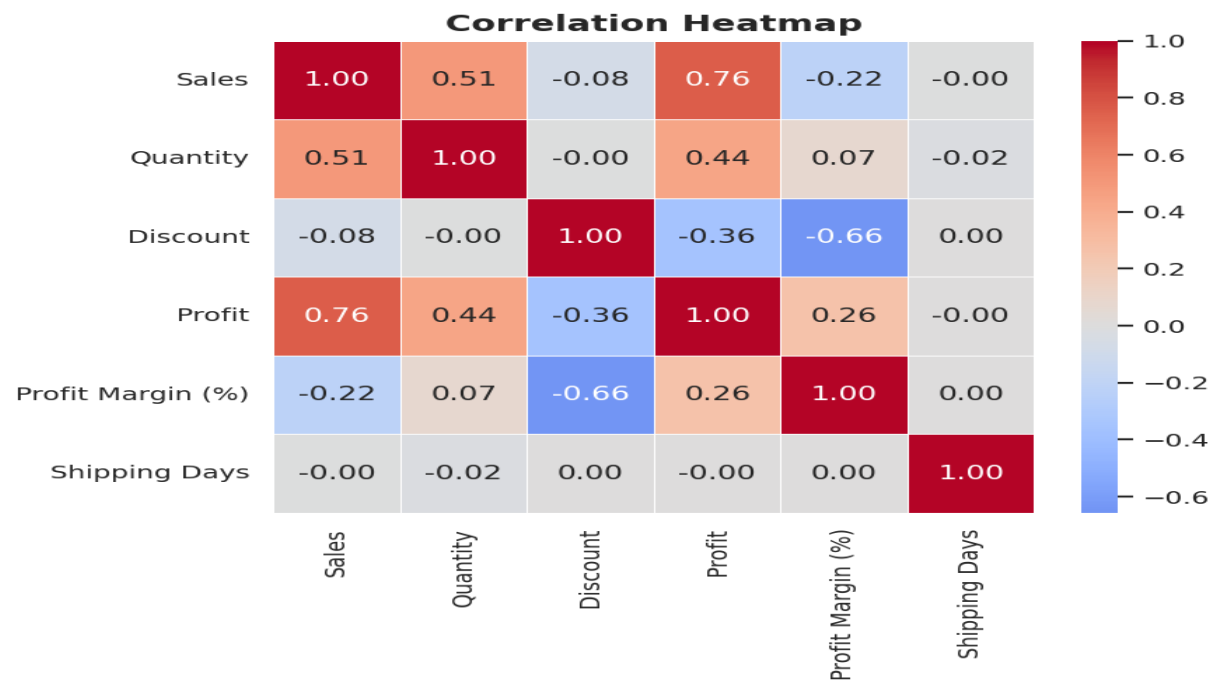
Insight: Top customers are spread across multiple segments, indicating healthy revenue diversification.

Discount vs Profit



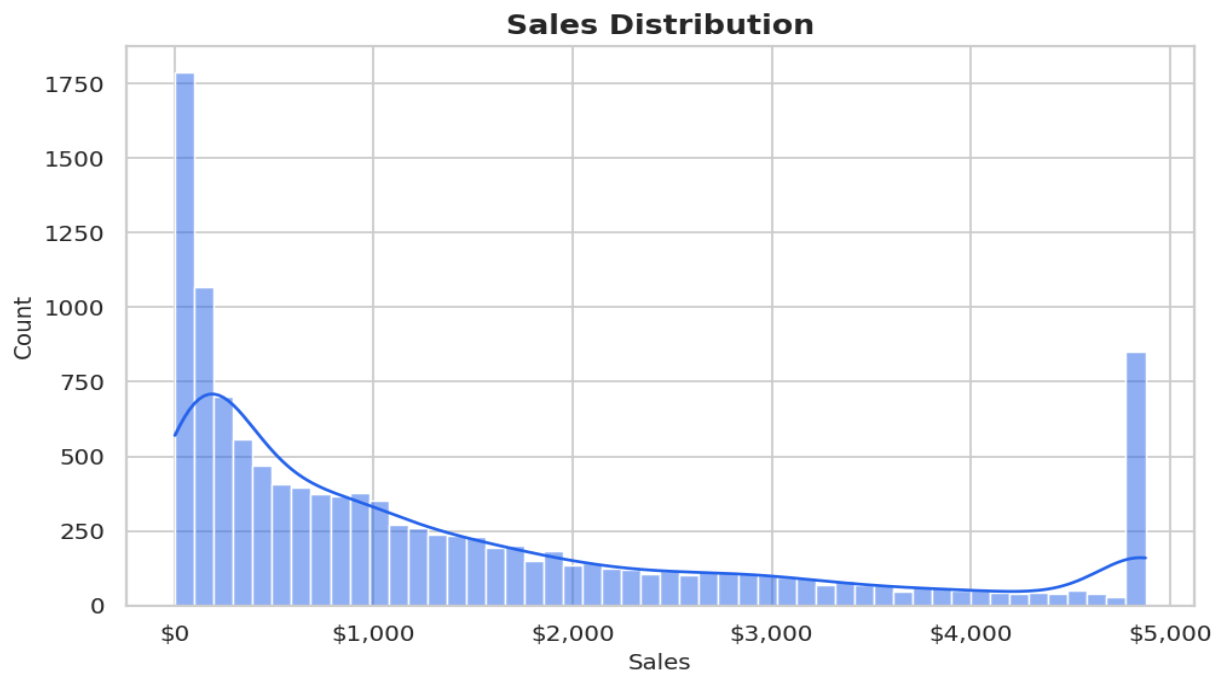
Insight: Profit deteriorates sharply beyond ~20% discount, with many high-discount orders becoming loss-making.

Correlation Heatmap



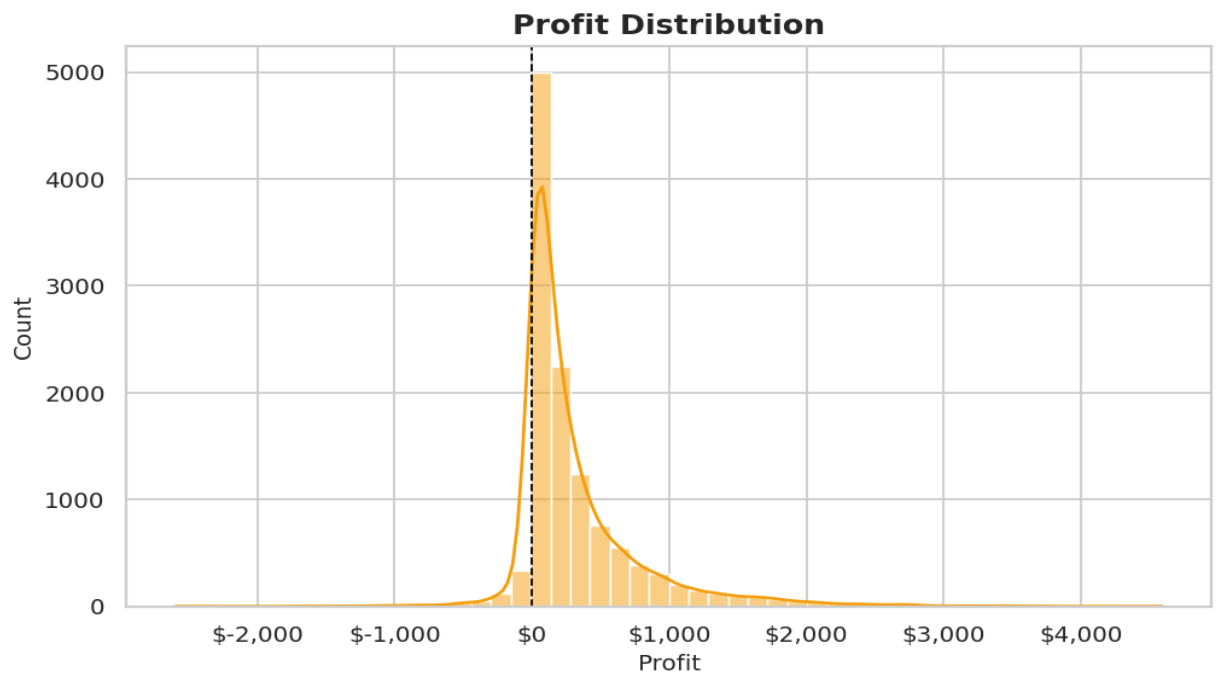
Insight: Discount is negatively correlated with Profit and Profit Margin; Shipping Days shows negligible correlation with financial outcomes.

Sales Distribution



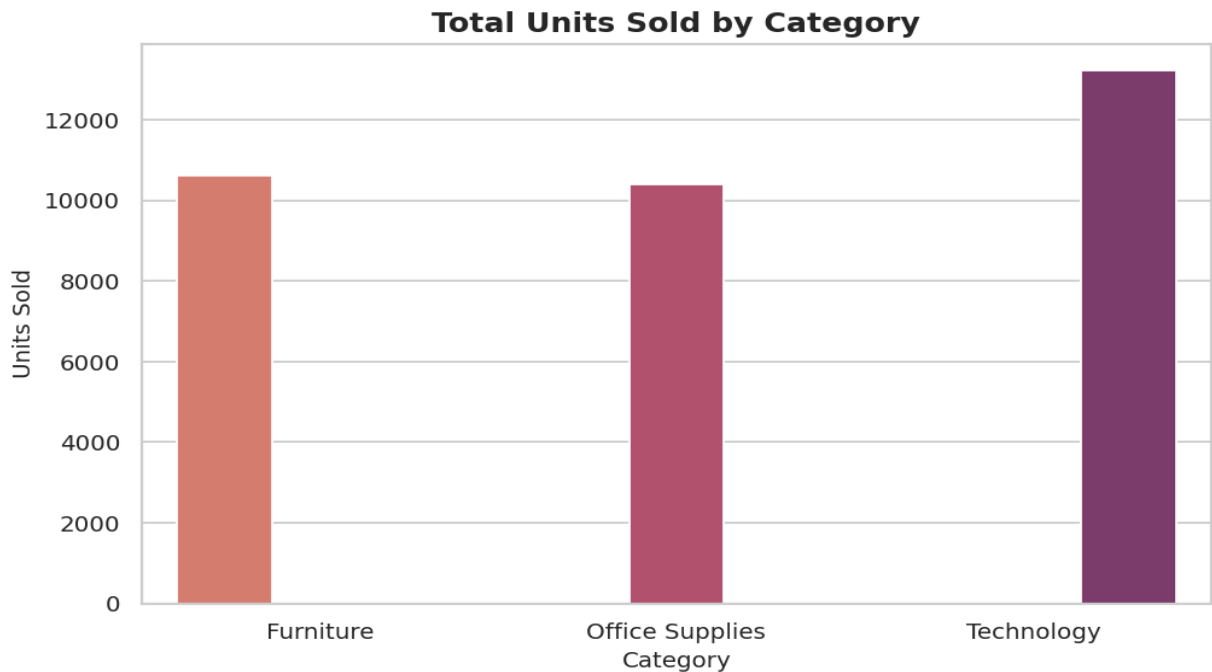
Insight: Sales values are right-skewed, with a long tail of high-value Technology transactions.

Profit Distribution



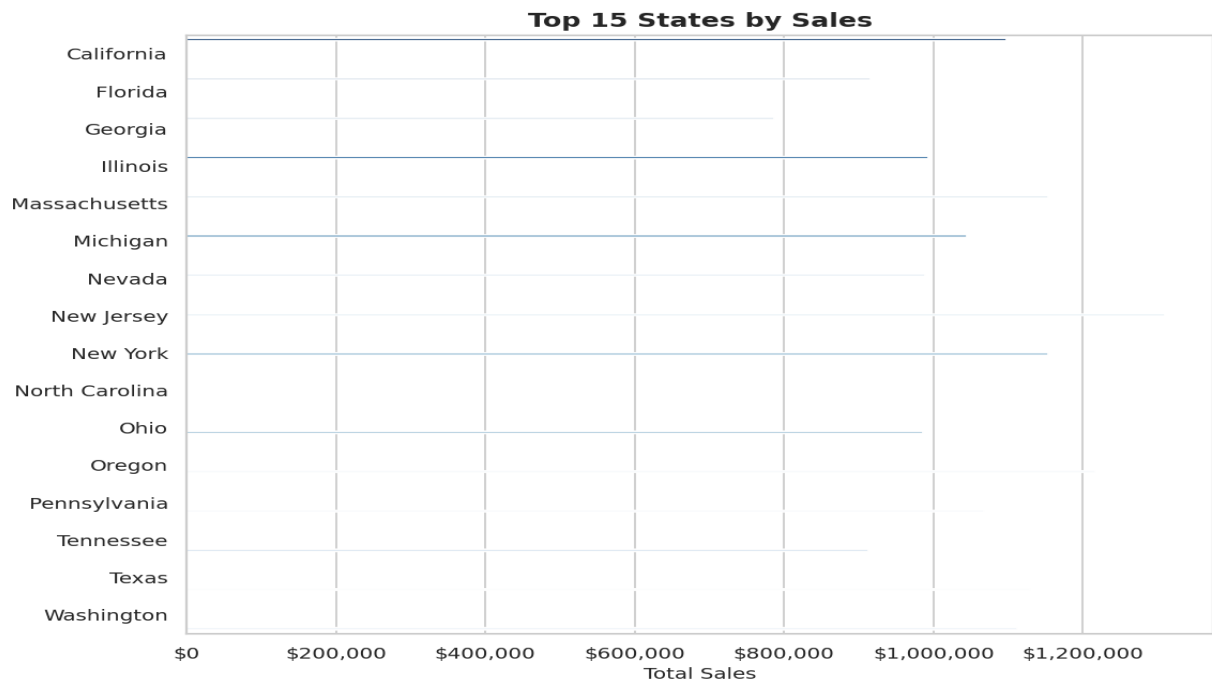
Insight: Profit is centered near a modest positive value with a visible left tail of loss-making orders.

Quantity Analysis by Category



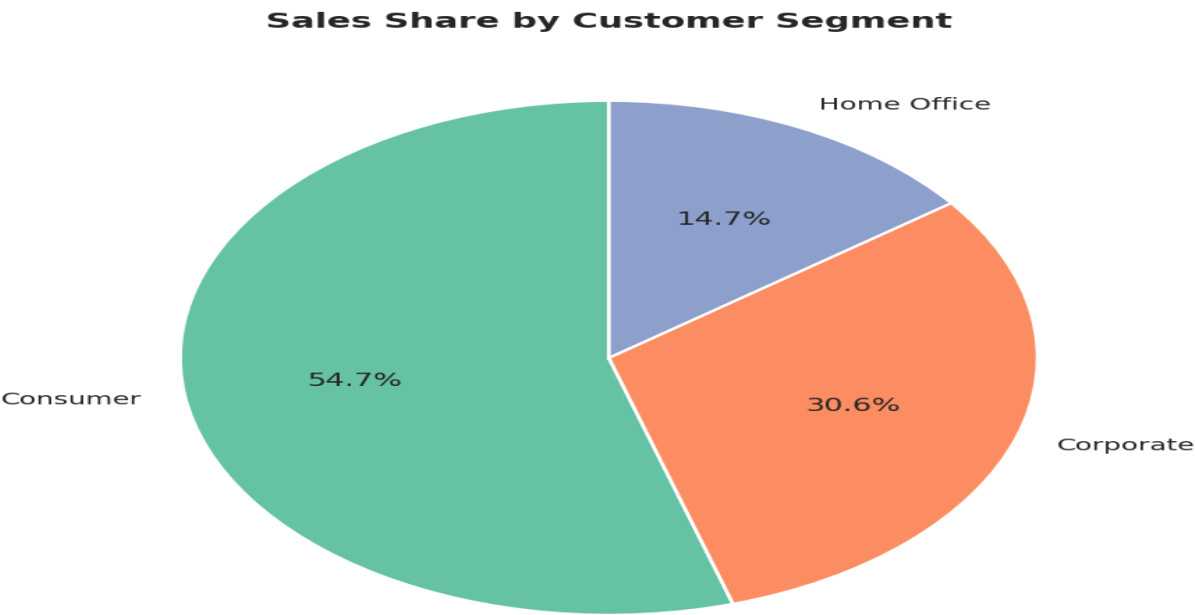
Insight: Office Supplies moves the highest unit volume despite generating the least revenue.

Top 15 States by Sales



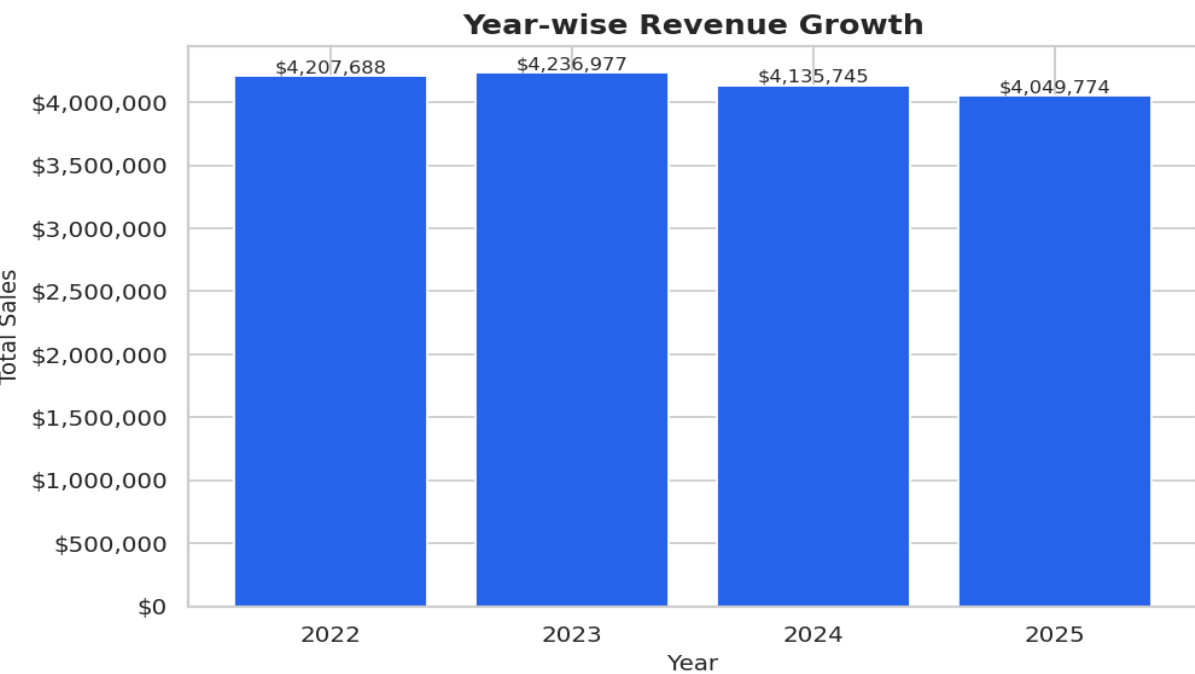
Insight: Sales concentrate in a handful of large states, highlighting both established markets and white-space opportunities.

Customer Segment Analysis



Insight: The Consumer segment contributes the largest revenue share, followed by Corporate and Home Office.

Year-wise Growth



Insight: Year-over-year growth is uneven, with at least one year showing a revenue decline that merits root-cause investigation.

Regional Performance Summary

Region	Total_Sales	Total_Profit	Orders	Avg_Discount
East	4684882.6	1074173.17	3336	0.09
West	4413858.31	979719.21	3252	0.09
Central	4152911.52	943944.21	2921	0.09
South	3378531.61	763024.35	2471	0.09

Customer Segment Summary

Segment	Total_Sales	Total_Profit	Orders
Consumer	9092705.45	2044529.96	6550
Corporate	5089982.52	1159018.39	3612
Home Office	2447496.06	557312.59	1818

Top 10 Products

Product Name	Total_Sales	Total_Profit	Units_Sold
OfficeCopier 4000	633143.12	128014.54	784
UltraSlim Laptop	623665.82	127656.45	812
Aria Smartphone	603437.33	108951.82	829
NovaPhone S8	596630.48	124030.72	742
Bluetooth Speaker	594360.34	116591.89	754
ProBook 15	589657.43	119484.75	746
ScanPro Duo	565627.48	110195.94	742
PrintMaster 200	560338.6	108831.26	716
Laptop Stand	558240.19	102155.54	743
GameStation Tower	553724.18	117653.18	743

Consolidated Business Insights

- 1 Technology drives the business, led by a small number of high-ticket products.
- 2 Discounting above ~20% erodes profitability and is the largest controllable margin lever.
- 3 The South region consistently underperforms the East, West, and Central regions.
- 4 Revenue is right-skewed — a small number of large orders drive a disproportionate share of total sales.
- 5 Office Supplies is a high-volume, low-value category, likely serving a retention/relationship role.
- 6 The Consumer segment leads in volume; Corporate leads in average order value.
- 7 Shipping speed shows negligible correlation with profit, so logistics is not the current constraint.
- 8 Year-over-year growth is uneven, with at least one year showing a decline that needs investigation.
- 9 Revenue is not dangerously concentrated in a single top customer.
- 10 Sales are geographically concentrated in a handful of large states, leaving mid-tier states as white space.

Actionable Recommendations

- 1 Cap discounts at 20% for Technology and Furniture orders unless manager-approved.
- 2 Introduce margin-protected discount tiers (0–10% self-serve, 10–20% approval, 20%+ escalation).
- 3 Launch a regional growth campaign in the South, mirroring East-region tactics.
- 4 Prioritize inventory and marketing spend on top-selling Technology products; consider bundling.
- 5 Reposition Office Supplies as a loyalty/retention category to drive repeat purchase frequency.
- 6 Build a dedicated Corporate account-management motion given its higher average order value.
- 7 Investigate the year with declining revenue to isolate pricing, mix, or market causes.
- 8 Expand marketing investment in mid-tier states showing moderate but growing sales.
- 9 Set up automated monthly profit-trend monitoring to catch loss-trending months early.
- 10 Review pricing on the lowest-margin sub-categories identified in the category summary.
- 11 Codify and share top sales representatives' playbooks across the team.
- 12 Introduce a loyalty program for high-value Consumer-segment customers.
- 13 Bundle slow-moving sub-categories with top sellers to increase sell-through without heavy discounting.
- 14 Test faster shipping options in lower-sales regions to rule out logistics as a hidden barrier.
- 15 Establish a quarterly business review (QBR) cadence using this report's KPIs to track initiative impact.

Conclusion

The business maintains a fundamentally healthy revenue base anchored by the Technology category, with the single largest controllable opportunity being tighter discount discipline. Regional performance also shows a clear and addressable gap, with the South region trailing the rest of the business. Acting on the recommendations in this report — particularly discount governance and regional investment — offers a realistic, low-risk path to improving both revenue growth and profit margin without requiring new product lines.

Future Improvements

- Incorporate marketing spend and customer acquisition cost data to compute true ROI by channel and region.
- Build a cohort-based customer lifetime value (CLV) model to prioritize retention investment.
- Add a forecasting layer (e.g., Prophet or ARIMA) to project next-quarter revenue and proactively flag risk months.
- Automate this report and the interactive dashboard to refresh on a scheduled basis as new transaction data arrives.